



Lecture 4 - Ethics for IT Workers and IT Users

Learning Objectives

After the completion of the chapter, the students will be able to:

- describe the key characteristics that distinguish a professional from other kinds of workers and as an IT worker considered a professional;
- explain the factors transforming the professional services industry; and
- discuss the relationship that an IT worker manages, and the key ethical issues can arise in each.

IT Worker Relationship That Must Be Managed

IT workers typically become involved in many different work relationships, including those with employers, clients, suppliers, other professionals, IT users, and society at large. In each relationship, an ethical IT worker act honestly and appropriately.

Relationships Between IT Workers and Employers

IT workers and employers have a critical, multifaceted relationship that requires ongoing effort by both parties to keep it strong. An IT worker and an employer typically agree on the fundamental aspects of this relationship before the worker accepts an employment offer.

Other aspects of this relationship develop over time, depending on circumstances. Some aspects are addressed by law. Some issues are specific to the role of the IT worker and are established based on the nature of the work or project.

Relationships Between IT Workers and Employers

As the stewards of an organization's IT resources, IT workers must set an example and enforce policies regarding the ethical use of IT. IT workers often have the skills and knowledge to abuse systems and data or to enable others to do so. Software piracy is an area in which IT workers may be tempted to violate laws and policies. Although end users often get the blame when it comes to using illegal copies of commercial software, software piracy in a corporate setting is sometimes directly traceable to IT staff members—either they allow it to happen or they actively engage in it, often to reduce IT-related spending.

Relationships Between IT Workers and Employers

Trade secrecy is another area that can present challenges for IT workers and their employers. A trade secret is information, generally unknown to the public, that a company has taken strong measures to keep confidential. It represents something of economic value that has required effort or cost to develop and that has some degree of uniqueness or novelty. Trade secrets can include the design of new software code, hardware designs, business plans, the design of a user interface to a computer program, and manufacturing processes.

Relationships Between IT Workers and Employers

Another issue that can create friction between employers and IT workers is whistleblowing. Whistle-blowing is an effort by an employee to attract attention to a negligent, illegal, unethical, abusive, or dangerous act by a company that threatens the public interest. Whistle-blowers often have special information based on their expertise or position within the offending organization.

Relationships Between IT Workers and Clients

IT workers provide services to clients; sometimes those “clients” are coworkers who are part of the same company as the IT worker. In other cases, the client is part of a different company. In relationships between IT workers and clients, each party agrees to provide something of value to the other. Generally speaking, the IT worker provides hardware, software, or services at a certain cost and within a given time frame.

Relationships Between IT Workers and Clients

Typically, the client makes decisions about a project on the basis of information, alternatives, and recommendations provided by the IT worker. The client trusts the IT worker to use his or her expertise and to act in the client's best interests. The IT worker must trust that the client will provide relevant information, listen to and understand what the IT worker says, ask questions to understand the impact of key decisions, and use the information to make wise choices among various alternatives. Thus, the responsibility for decision making is shared between the client and the IT worker.

Relationships Between IT Workers and Clients

One potential ethical problem that can interfere with the relationship between IT workers and their clients involves IT consultants or auditors who recommend their own products and services or those of an affiliated vendor to remedy a problem they have detected. Such a situation has the potential to undermine the objectivity of an IT worker due to a conflict of interest—a conflict between the IT worker's (or the IT firm's) self-interest and the client's interests

Relationships Between IT Workers and Clients

Problems can also arise during a project if IT workers find themselves unable to provide full and accurate reporting of the project's status due to a lack of information, tools, or experience needed to perform an accurate assessment. The project manager may want to keep resources flowing into the project and hope that problems can be corrected before anyone notices. The project manager may also be reluctant to share status information because of contractual penalties for failure to meet the schedule or to develop certain system functions.

Relationships Between IT Workers and Clients

Fraud is the crime of obtaining goods, services, or property through deception or trickery. Fraudulent misrepresentation occurs when a person consciously decides to induce another person to rely and act on a misrepresentation. To prove fraud in a court of law, prosecutors must demonstrate the following elements:

- The wrongdoer made a false representation of material fact.
- The wrongdoer intended to deceive the innocent party.
- The innocent party justifiably relied on the misrepresentation.
- The innocent party was injured.

Relationships Between IT Workers and Clients

Misrepresentation is the misstatement or incomplete statement of a material fact. If the misrepresentation causes the other party to enter into a contract, that party may have the legal right to cancel the contract or seek reimbursement for damages.

Relationships Between IT Workers and Clients

Breach of contract occurs when one party fails to meet the terms of a contract. Further, a material breach of contract occurs when a party fails to perform certain express or implied obligations, which impairs or destroys the essence of the contract. Because there is no clear line between a minor breach and a material breach, determination is made on a case-by-case basis. “When there has been a material breach of contract, the non-breaching party can either: (1) cancel the contract, seek restitution of any compensation paid under the contract to the breaching party, and be discharged from any further performance under the contract; or (2) treat the contract as being in effect and sue the breaching party to recover damages.

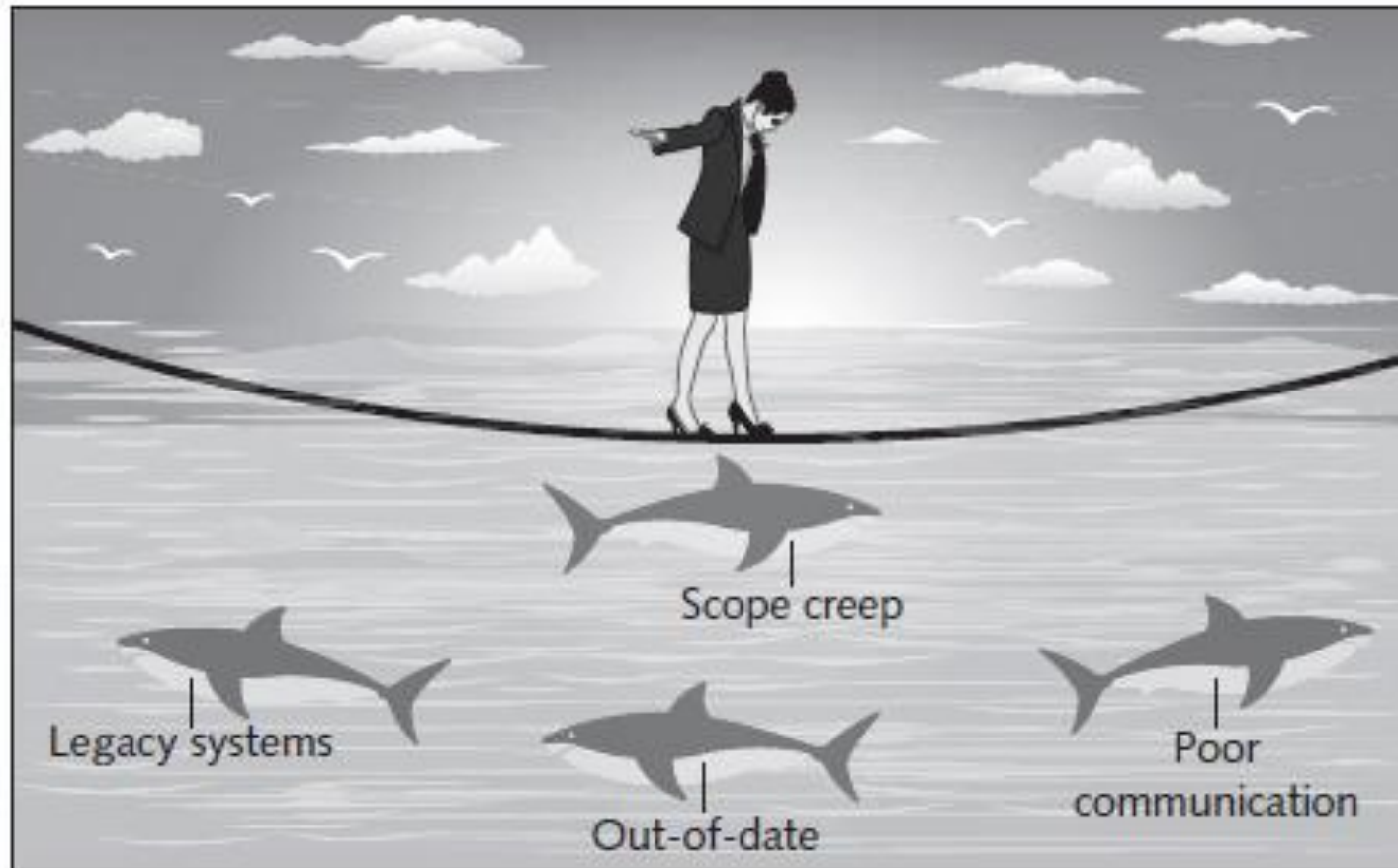
Relationships Between IT Workers and Clients

Frequent causes of problems in IT projects include the following:

- Scope creep
- Poor communication
- Delivery of an obsolete solution
- Legacy systems

Relationships Between IT Workers and Clients

Frequent Causes of Problems in IT Projects



Relationships Between IT Workers and Suppliers

IT workers deal with many different hardware, software, and service providers. Most IT workers understand that building a good working relationship with suppliers encourages the flow of useful communication as well as the sharing of ideas. Such information can lead to innovative and cost-effective ways of using the supplier's products and services that the IT worker may never have considered.

Suppliers strive to maintain positive relationships with their customers in order to make and increase sales. To achieve this goal, they may sometimes engage in unethical actions—for example, offering an IT worker a gift that is actually intended as a bribe. Clearly, IT workers should not accept a bribe from a vendor, and they must be careful when considering what constitutes a bribe.

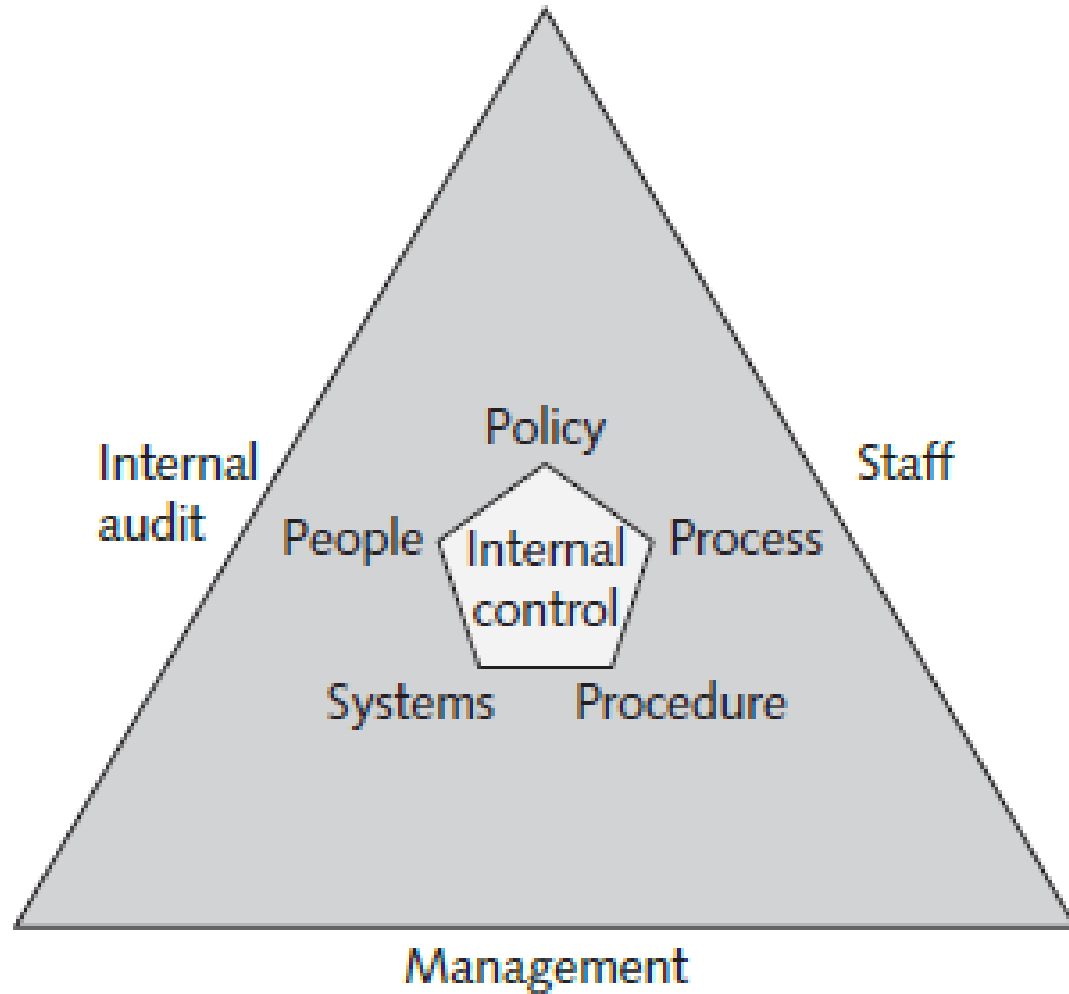
Relationships Between IT Workers and Suppliers

Bribery is the act of providing money, property, or favors to someone in business or government in order to obtain a business advantage. An obvious example is a software supplier sales representative who offers money to another company's employee to get its business. This type of bribe is often referred to as a kickback or a payoff. The person who offers a bribe commits a crime when the offer is made, and the recipient is guilty of a crime if he or she accepts the bribe.

Relationships Between IT Workers and Suppliers

Internal control is the process established by an organization's board of directors, managers, and IT systems people to provide reasonable assurance for the effectiveness and efficiency of operations, the reliability of financial reporting, and compliance with applicable laws and regulations. An organization's internal control resources include all the people, policies, processes, procedures, and systems controlled by management that enable it to meet these goals

Relationships Between IT Workers and Clients



Internal Control

Relationships Between IT Workers and Suppliers

Policies are the guidelines and standards by which the organization must abide. The guidelines and standards are often in response to some law. Policies drive processes and procedures. **Processes** are a collection of tasks designed to accomplish a stated objective. A **procedure** defines the exact instructions for completing each task in a process.

Management is responsible for ensuring that an adequate system of internal control is set up, documented with written procedures, and implemented. Management must also decide the proper level of control over various aspects of the business so that the cost of implementing control does not outweigh the benefits.

Relationships Between IT Workers and Suppliers

In some countries, gifts are an essential part of doing business. In fact, in some countries, it would be considered rude not to bring a present to an initial business meeting. In some countries, a gift might take the form of free tickets to a sporting event from a personnel agency that wants to get on your company's list of preferred suppliers.

Relationships Between IT Workers and Suppliers

Distinguishing between bribes and gifts

Bribes	Gifts
Are made in secret, as they are neither legally nor morally acceptable	Are made openly and publicly, as a gesture of friendship or goodwill
Are often made indirectly through a third party	Are made directly from donor to recipient
Encourage an obligation for the recipient to act favorably toward the donor	Come with no expectation of a future favor for the donor

Relationships Between IT Workers and Other Professionals

Professionals often feel a degree of loyalty to the other members of their profession. As a result, they are often quick to help each other obtain new positions but slow to criticize each other in public. Professionals also have an interest in their profession as a whole, because how it is perceived affects how individual members are viewed and treated.

Relationships Between IT Workers and Other Professionals

Area of Exaggeration	Frequency (%)	How to uncover the truth
Overstated skill set	57	Verification of licenses and/or certifications with accrediting agency
Embroidered responsibilities	55	Thorough background and reference checks
Dates of employment	42	Thorough background and reference checks
Job title	34	Thorough background and reference checks
Academic degrees earned	33	Verification of education claims with educational institutions
Companies worked for	26	Thorough background and reference checks
Awards	18	Thorough background and reference checks

Relationships Between IT Workers and Other Professionals

Another ethical issue that can arise in relationships between IT workers and other professionals is the inappropriate sharing of corporate information. Because of their roles, IT workers may have access to corporate databases of private and confidential information about employees, customers, suppliers, new product plans, promotions, budgets, and so on. It might be sold to other organizations or shared informally during work conversations with others who have no need to know. Revealing such private or confidential information is grounds for termination in many organizations and could even lead to criminal charges.

Relationships Between IT Workers and IT Users

IT user - refers to a person who uses a hardware or software product

- distinguishes end users from the IT workers who develop, install, service, and support the product.
- have a duty to understand a user's needs and capabilities and to deliver products and services that best meet those needs—subject, of course, to budget and time constraints.
- have a key responsibility to establish an environment that supports ethical behaviors by users

Relationships Between IT Workers and Society

Laws establish safety standards for products and services to protect the public. However, these laws are less than perfect, and they cannot safeguard against all negative side effects of a product or process. Often, professionals can clearly see the effect their work will have and can take action to eliminate potential public risks.

Professional Codes of Ethics

A **professional code of ethics** states the principles and core values that are essential to the work of a particular occupational group. Practitioners in many professions subscribe to a code of ethics that governs their behavior. Most codes of ethics created by professional organizations have **two main parts**: The first outlines **what the organization aspires to become** and the second typically **lists rules and principles by which members of the organization are expected to abide**. Many codes also include a commitment to continuing education for those who practice the profession.

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Professional Codes of Ethics

Benefits:

- ✓ Ethical decision making
- ✓ High standards of practice and ethical behavior
- ✓ Trust and respect from the general public
- ✓ Evaluation benchmark

Professional Organizations

- Association for Computing Machinery (ACM)
- SysAdmin, Audit, Network, Security (SANS) Institute
- Institute of Electrical and Electronics Engineers Computer Society (IEEE-CS)
- Association of Information Technology Professionals (AITP)

Certification

Certification indicates that a professional possesses a particular set of skills, knowledge, or abilities, in the opinion of the certifying organization. Unlike licensing, which applies only to people and is required by law, certification can also apply to products and is generally voluntary. IT-related certifications may or may not include a requirement to adhere to a code of ethics, whereas such a requirement is standard with licensing.

Certification

Common vendor-specific certifications for IT workers

Category	Certification
MAC OS X	Apple Certified Technical Coordinator
Cisco Hardware	Cisco Certified Design Associate
Cisco Networking	Cisco Certified Network Professionals
Cisco Networking	Cisco Certified Internetwork Expert
Microsoft Products	Microsoft Certified Professional
Citrix Products	Citrix Certified Administrator (CCA)
Oracle Database	Oracle Database 12c: Certified Expert Performance Management and Tuning
Salesforce software	Salesforce.com Certified Administrator

Licensing of IT Professionals

A **government license** is government-issued permission to engage in an activity or to operate a business. Most license activities that could result in damage to public health, safety, or welfare—if practiced by an individual who has not demonstrated minimal competence. Licensing is generally administered and often requires that the recipient pass a test of some kind.

The Case of Licensing IT Workers

Software Engineering Core of Ethics and Professional Practice

Software engineers shall commit themselves to making the analysis, specification, design, development, testing and maintenance of software a beneficial and respected profession. In accordance with their commitment to the health, safety and welfare of the public, software engineers shall adhere to the following Eight Principles:

1. Public - Software engineers shall act consistently with the public interest.
2. Client and Employer - Software engineers shall act in a manner that is in the best interests of their client and employer consistent with the public interest.
3. Product - Software engineers shall ensure that their products and related modifications meet the highest professional standards possible.
4. Judgment - Software engineers shall maintain integrity and independence in their professional judgment.
5. Management - Software engineering managers and leaders shall subscribe to and promote an ethical approach to the management of software development and maintenance.
6. Profession - Software engineers shall advance the integrity and reputation of the profession consistent with the public interest.
7. Colleagues - Software engineers shall be fair to and supportive of their colleagues.
8. Self - Software engineers shall participate in lifelong learning regarding the practice of their profession and shall promote an ethical approach to the practice of the profession.

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IT Professional Malpractice

Negligence is defined as not doing something that a reasonable person would do or doing something that a reasonable person would not do. Duty of care refers to the obligation to protect people against any unreasonable harm or risk.

If a court finds that a defendant actually owed a duty of care, it must then determine whether the duty was breached. A **breach of the duty of care** is the failure to act as a reasonable person would act.

Professionals who breach the duty of care are liable for injuries that their negligence causes. This liability is commonly referred to as **professional malpractice**

Common Ethical Issues for IT Users

- Software Piracy
- Inappropriate Use of Computing Resources
- Inappropriate Sharing of Information

Supporting the Ethical Practices of IT Users

- Establishing Guidelines for Use of Company Hardware and Software
- Defining an Acceptable Use Policy (AUP)
 - a document that stipulates restrictions and practices that a user must agree to in order to use organizational computing and network resources

5 Key Elements

1. Purpose of the AUP
2. Scope
3. Policy
4. Compliance
5. Sanctions

Supporting the Ethical Practices of IT Users

Question	Yes	No
Is there a statement that explains the need for an acceptable use policy?		
Is it clear how the policy applies to the following types of workers?		
• Full-time employees • Part-time employees • Temps • Contractors		
Does the policy address the following issues?		
• Protection of the data privacy rights of employees, customers, suppliers, and others • Control of access to proprietary company data and information • Use of unauthorized or pirated software • Employee monitoring, including email, wiretapping and eavesdropping on phone conversations, computer monitoring, and surveillance by video • Respect of the intellectual rights of others, including trade secrets, copyrights, patents, and trademarks • Inappropriate use of IT resources, such as web surfing, excessive use of social networks, blogging, personal emailing, and other use of computers for purposes other than business • The need to protect the security of IT resources through adherence to good security practices, such as not sharing user IDs and passwords, using hard-to-guess passwords, and frequently changing passwords • The use of the computer to intimidate, harass, or insult others through abusive language in emails and by other means		
Are disciplinary actions defined for IT-related abuses?		
Is there a process for communicating the policy to employees?		
Is there a plan to provide effective, ongoing training relative to the policy?		

Manager's checklist for establishing an acceptable use policy

Supporting the Ethical Practices of IT Users

- Structuring Information Systems to Protect Data and Information
- Installing and Maintaining a Corporate Firewall

Firewall - hardware or software (or a combination of both) that serves as the first line of defense between an organization's network and the Internet

- limits access to the company's network based on the organization's Internet-usage policy.
- can be configured to serve as an effective deterrent to unauthorized web surfing by blocking access to specific objectionable websites.
- Compliance